Little Lab Coats

🦎 Nature's Engineers: How Animal Bodies Solve Problems 🦆

Life Science Unit, Lesson 1

📚 **Grade:** 1st Grade ⏱️ **Duration:** 45 minutes 🔬 **Subject:** Life Science — Animal Structures 🧰 **Kit Lesson:** No

Nature has been solving engineering problems for billions of years — can you figure out how?

📋 **NGSS Standard:**
1-LS1-1 — Use materials to design a solution to a human problem by mimicking how plants and/or animals use their external parts to help them survive, grow, and meet their needs.

🎯 Learning Objective

Students will **identify** how specific animal body parts solve survival problems and **design** a simple invention inspired by one animal's structure.

💡 The Big Idea

Engineers often get their best ideas from nature! When animals evolve a clever body part — hooked burrs for traveling, webbed feet for swimming, hollow bird bones for flying — human engineers have co[illegible] these ideas to solve real problems. This is called **biomimicry** (bio = life, mimicry = copying).

🔍 **Did you know?** In 1941, a Swiss engineer named George de Mestral went on a walk with his dog, Milka. When they got home, burrs were stuck all over the dog's fur — and on his own jacket! He looked at them under a microscope and saw tiny hooks. Eight years later, he invented Velcro — copying nature's design. Look at the Velcro on your shoes or backpack and you'll see those same tiny hooks.

🔤 **Key vocabulary:** structure, function, biomimicry, engineer, adapt, survive, tool

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🌱 Roots, Stems, Leaves, Flowers — What Each Part Does 🌸

Life Science Unit 1, Lesson 2

📚 **Grade:** 1st Grade ⏱️ **Duration:** 45 minutes 🔬 **Subject:** Life Science — Plant Structures 📖 **Lesson:** 2 of 6

Every part of a plant has a job — and plants figured out how to survive without a mouth, stomach, or legs!

📋 **NGSS Standard:**
1-LS1-1 — Use materials to design a solution to a human problem by mimicking how plants and/or animals use their external parts to help them survive, grow, and meet their needs.

🎯 Learning Objective

Students will **identify** the four main plant parts and **explain** the survival function of each part.

💡 The Big Idea

Plants are survival experts — but they do everything differently than animals. They can't chase food, run from danger, or drink from a stream. Instead, each plant part is perfectly designed for a specific job. Together, these parts help the plant get water, make food, and make more plants.

💡 **Key vocabulary:** roots, stem, leaves, flower, photosynthesis, nutrients, absorb, pollinator, reproduce

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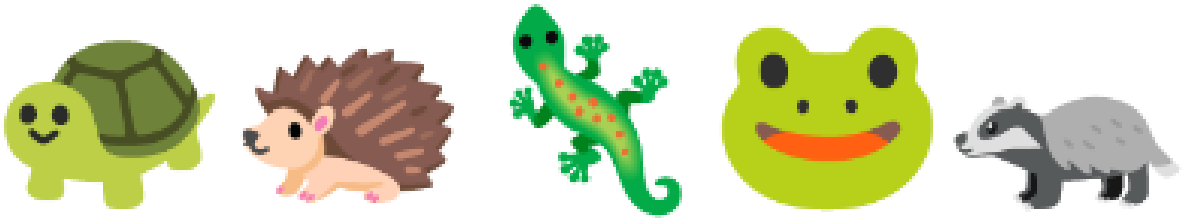
🛡️ How Animals Protect Themselves 🦔

Life Science Unit 1, Lesson 3

📚 **Grade:** 1st Grade ⏱️ **Duration:** 45 minutes 🔬 **Subject:** Life Science — Animal Structures 📖 **Lesson:** 3 of 6



Every animal has its own way of staying safe — and nature has invented some incredible solutions!

📋 **NGSS Standard:**
1-LS1-1 — Use materials to design a solution to a human problem by mimicking how plants and/or animals use their external parts to help them survive, grow, and meet their needs.

🎯 Learning Objective

Students will **compare** at least four different structural defenses animals use and **design** a protection structure inspired by one.

💡 The Big Idea

In nature, being eaten is a very real problem. Animals have evolved amazing body structures — called **structural defenses** — to help them avoid predators. These structures are built right into the animal's body and don't require any learning or planning. Let's explore five clever defense strategies!

💡 **Key vocabulary:** defense, predator, prey, camouflage, armor, mimicry, spine, shell, structure

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🌸 How Plants Attract Pollinators 🐝

Life Science Unit 1, Lesson 4

📚 **Grade:** 1st Grade ⏱️ **Duration:** 45 minutes 🔬 **Subject:** Life Science — Plant Structures 📖 **Lesson:** 4 of 6

Flowers are nature's billboards — every color, shape, and smell is an advertisement aimed at a specific visitor!

📋 **NGSS Standard:**
1-LS1-1 — Use materials to design a solution to a human problem by mimicking how plants and/or animals use their external parts to help them survive, grow, and meet their needs.

🎯 Learning Objective

Students will **explain** how specific plant structures — color, smell, nectar, and shape — attract specific pollinators, and **design** their own "perfect flower" matched to a pollinator.

💡 The Big Idea

Flowers don't bloom to be beautiful for us — every feature of a flower evolved to attract a specific pollinator. Color, shape, scent, and the location of nectar are all structural features designed (by evolution) to lure exactly the right animal. When that animal visits, it picks up pollen and carries it to another flower — and new seeds can form. It's a partnership: the plant gets pollinated, the visitor gets food.

🔑 **Key vocabulary:** pollinator, pollen, nectar, pollination, structure, attract, reproduce, seed

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🦅 Animal Structures for Getting Food 🐘

Life Science Unit 1, Lesson 5

📚 **Grade:** 1st Grade ⏱️ **Duration:** 45 minutes 🔬 **Subject:** Life Science — Animal Structures 📖 **Lesson:** 5 of 6

Every animal has a specialized tool kit for getting food — no two are exactly alike!

📋 **NGSS Standard:**
1-LS1-1 — Use materials to design a solution to a human problem by mimicking how plants and/or animals use their external parts to help them survive, grow, and meet their needs.

🎯 Learning Objective

Students will **match** animal body structures to their food-getting function and **explain** why each structure suits that specific diet.

💡 The Big Idea

An animal's body parts are perfectly matched to what it eats. A meat-eater has different tools than a plant-eater. A filter-feeder has different tools than a seed-cracker. Over millions of years, animals with the best-matched body parts for their food source survived and passed on those features. The result: an amazing variety of specialized eating tools — beaks, tongues, trunks, claws, bills, and more!

💡 **Key vocabulary:** beak, talon, trunk, filter, prehensile, specialized, diet, structure, predator, prey

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🏃 Structures for Moving 🐠

Life Science Unit 1, Lesson 6 — Unit Finale!

📚 **Grade:** 1st Grade ⏱️ **Duration:** 45 minutes 🔬 **Subject:** Life Science — Animal Structures 📖 **Lesson:** 6 of 6 ⭐

Water, air, underground, treetops, desert — every environment has an animal perfectly built to move through it!

📋 **NGSS Standard:**
1-LS1-1 — Use materials to design a solution to a human problem by mimicking how plants and/or animals use their external parts to help them survive, grow, and meet their needs.

🎯 Learning Objective

Students will **identify** structural adaptations for movement in different environments and **design** an original animal with structures suited to a specific environment.

💡 The Big Idea

Moving around is essential for survival — animals need to find food, escape predators, find mates, and reach shelter. Different environments demand completely different movement strategies. The structures that make a fish an effortless swimmer would be useless underground. The paws that make a mole a perfect digger would slow it down in water. Every environment shapes its animals to move through it efficiently.

💡 **Key vocabulary:** streamlined, fins, membrane, adaptation, environment, burrow, prehensile, splayed, tundra, habitat

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